

## Week 4: Inclass Assignment

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### Dataset Name & File Used

- **Dataset:** SNAP Facebook Social Circles
- **File:** facebook\_combined.txt (converted to facebook\_edges.csv)

### Network Metrics

- **Nodes:** 4,039
- **Edges:** 88,234
- **Graph Type:** Undirected

### Top 5 Nodes by Degree

1. **Node 107:** 1,045 edges
2. **Node 1684:** 792 edges
3. **Node 1912:** 755 edges
4. **Node 3437:** 547 edges
5. **Node 0:** 347 edges

### Top 5 Nodes by Betweenness Centrality

1. Node 107
2. Node 1684
3. Node 567
4. Node 598
5. Node 4031

### Short Interpretation of Network Structure

The network follows a **Scale-Free** distribution, characterized by a few "super-hubs" (like Node 107) and many poorly connected nodes. The high **Clustering Coefficient** suggests that the network is composed of "Social Circles" groups where if A knows B and B knows C, A likely knows C as well. The presence of clear community structures (detected via the Modularity algorithm) indicates that Facebook users naturally organize into dense, localized clusters (e.g., schools, workplaces) connected to the wider network by a few key individuals.

### **What I Learned**

Through this analysis, I learned that social networks are not random. They are organized into "Small Worlds" where anyone can reach anyone else in a few steps because of high-degree hubs. I also learned that **Betweenness Centrality** is often more important than **Degree** for identifying influencers; a person might not have the most friends, but if they connect two different communities, they hold the most social power.